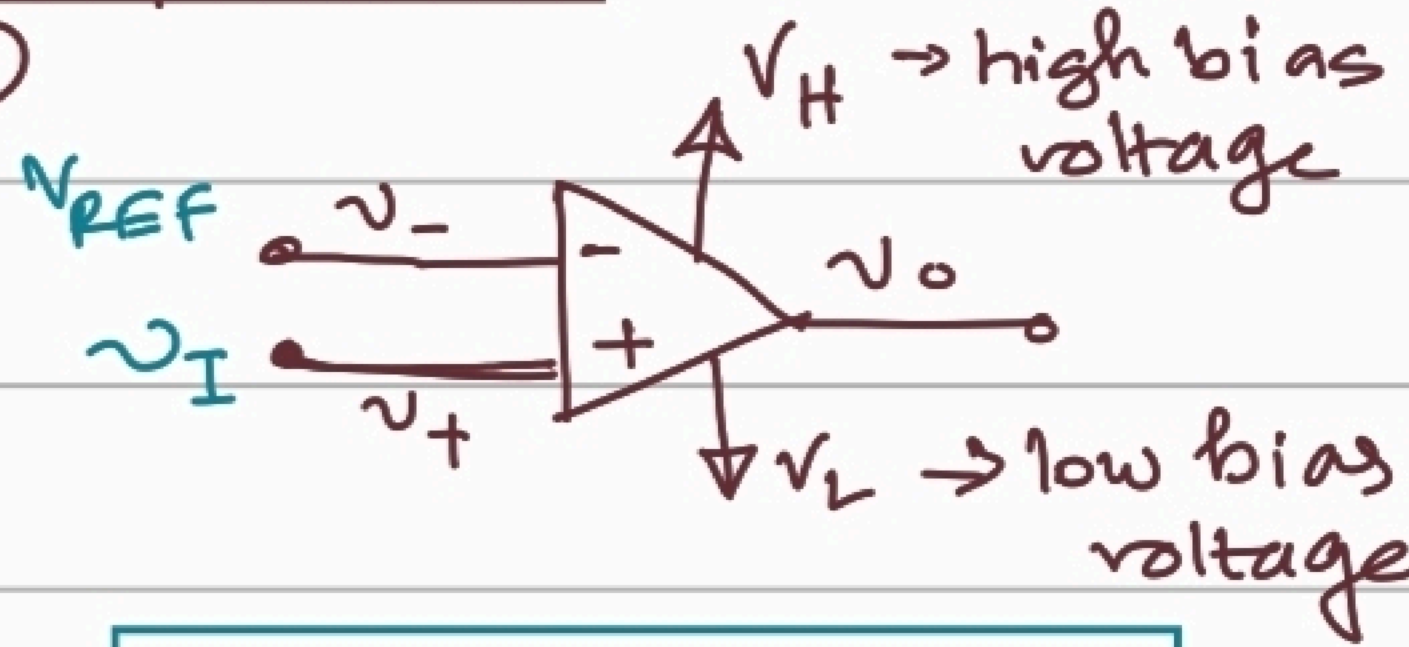
Lecture 18: Schmitt Trigger Circuit

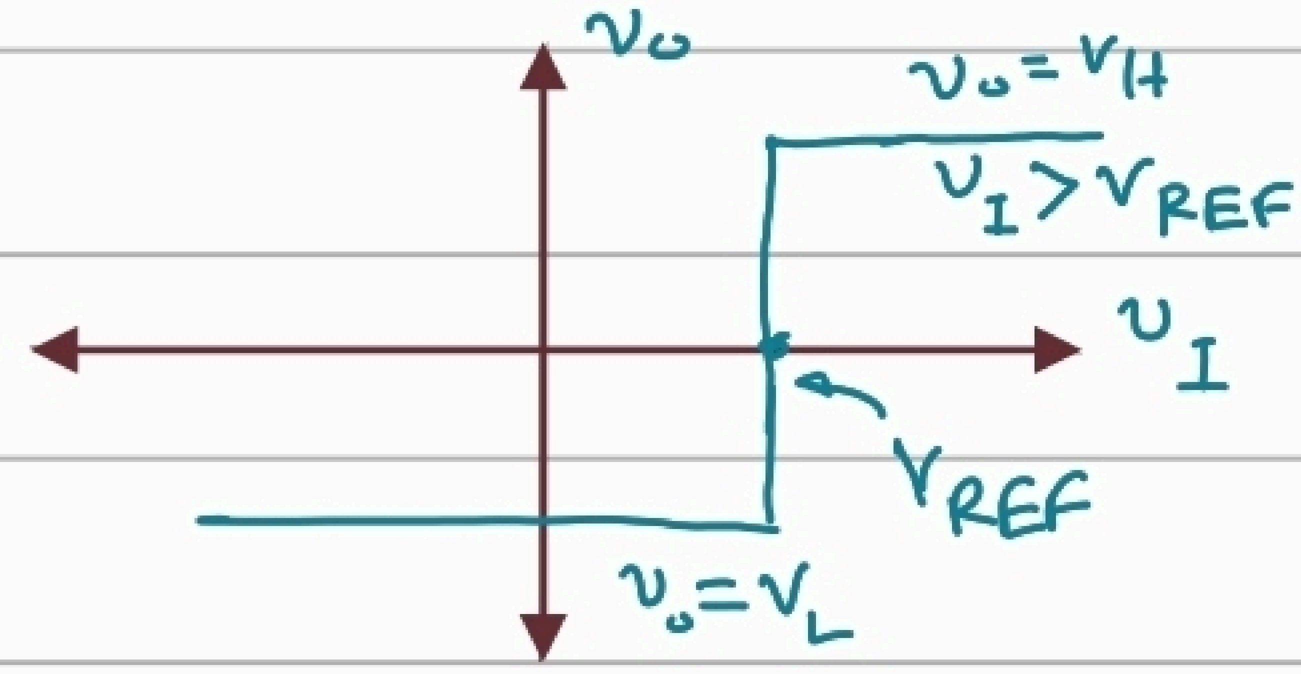
Otto H. Schmitt (1934) envisioned this circuit.

Comparators:

①



Transfer characteristics

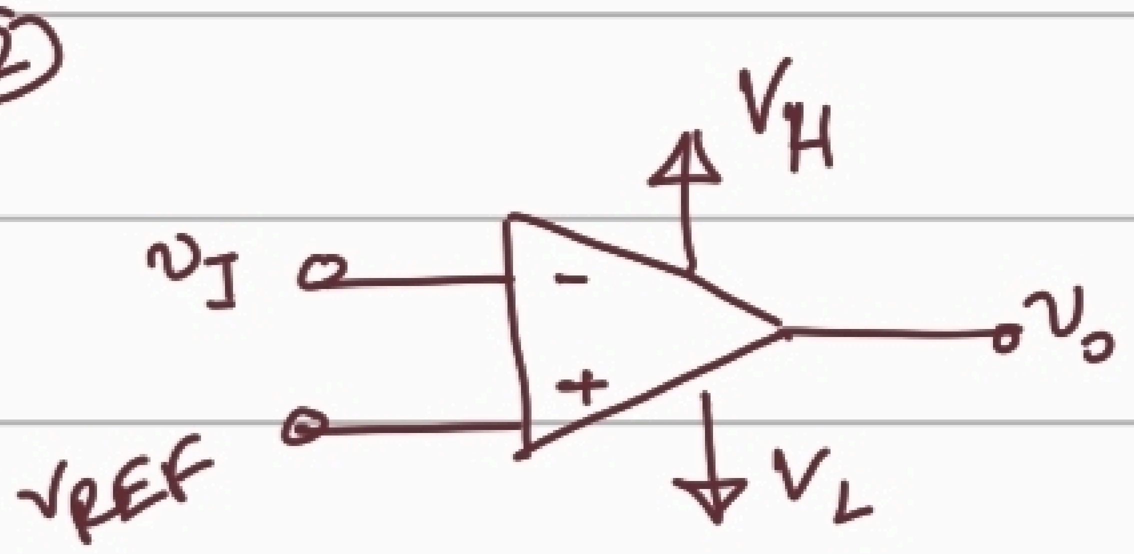


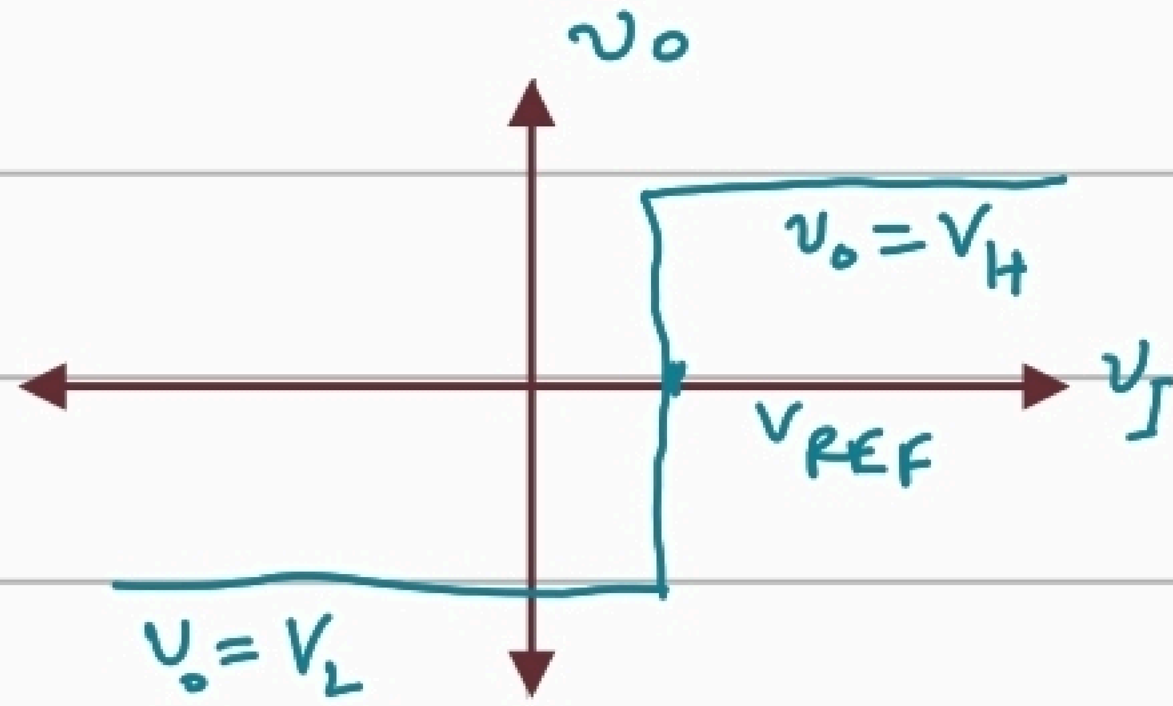
$$\boxed{\begin{array}{l} v_+ > v_- \Rightarrow v_o = V_H \\ v_+ < v_- \Rightarrow v_o = V_L \end{array}}$$

$v_+ = v_I > V_{REF} = v_- \Rightarrow v_o = V_H$

$v_+ = v_I < V_{REF} = v_- \Rightarrow v_o = V_L$

②

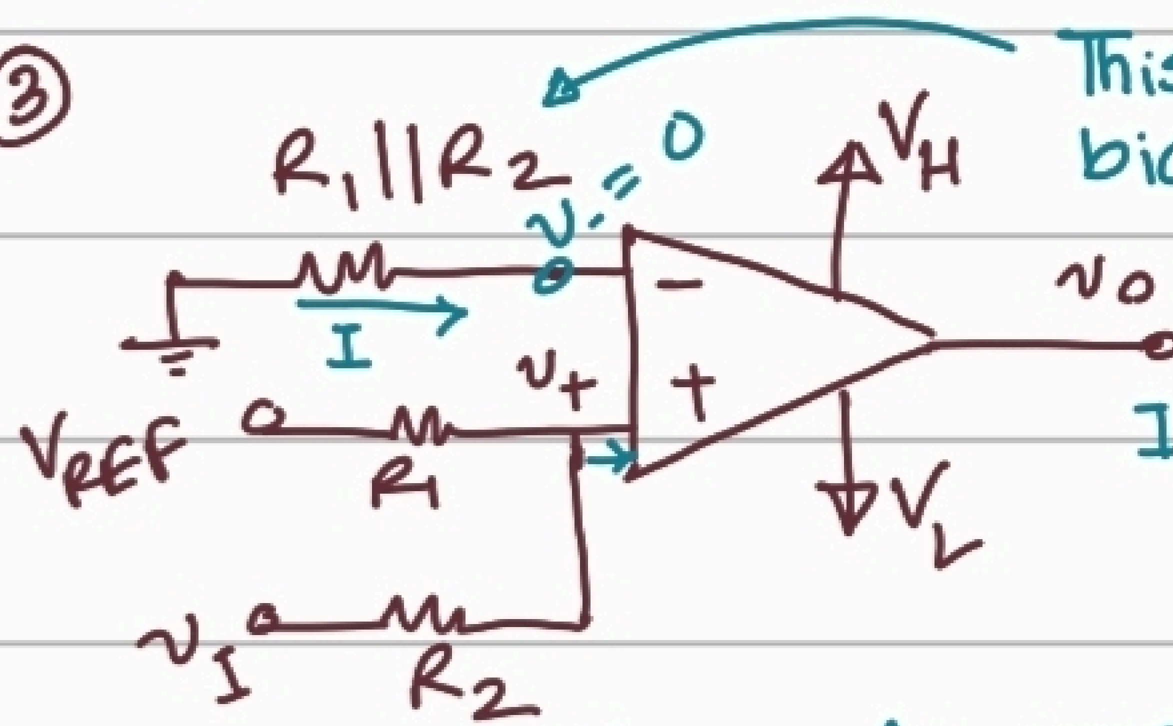




$v_+ = V_{REF} > v_I = v_- \Rightarrow v_o = V_H$

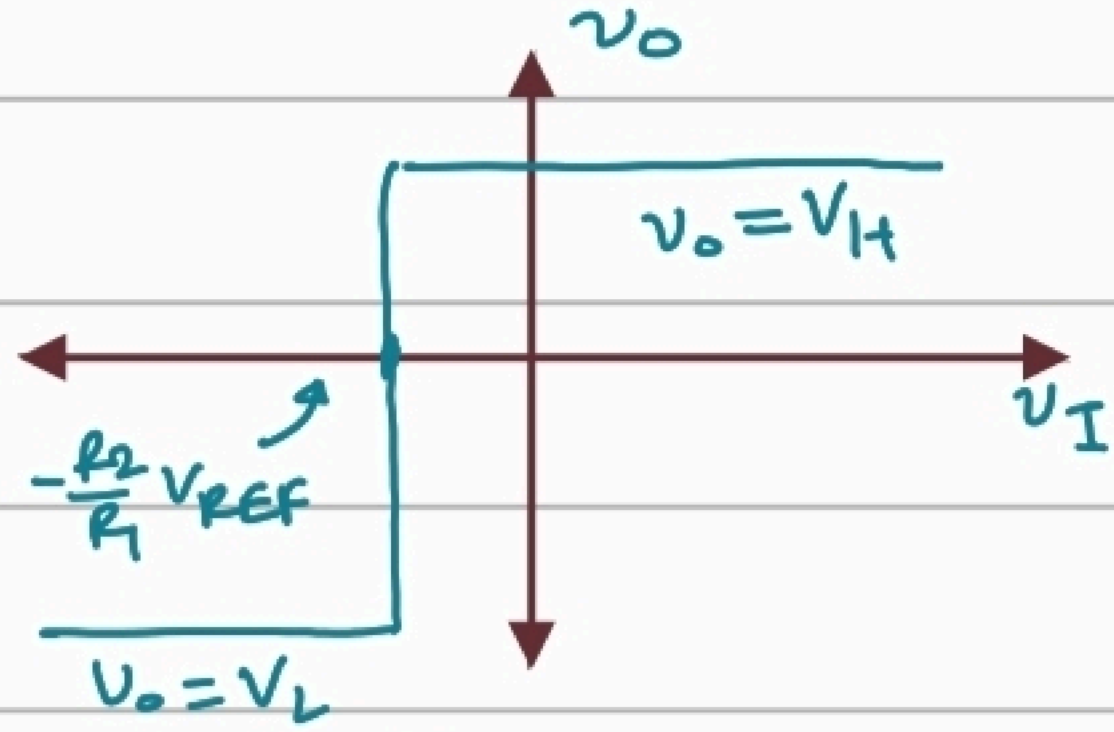
$v_+ = V_{REF} < v_I = v_- \Rightarrow v_o = V_L$

③

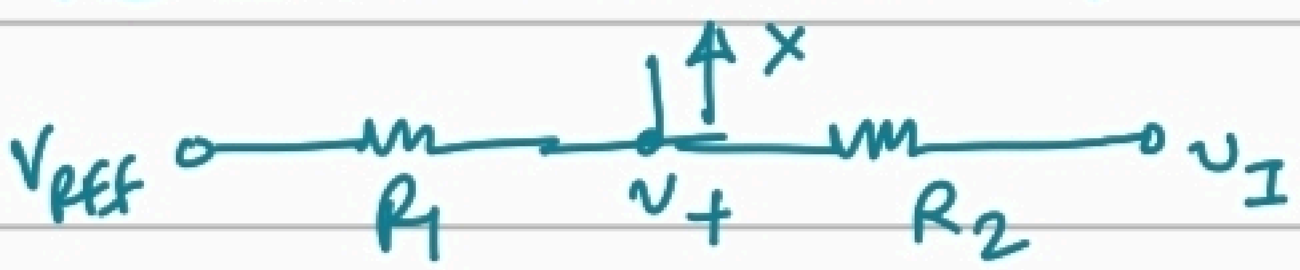


This prevent DC bias effect inside op-amp.

$I = 0$, $v_- = 0$ Always.



We can write node equation at "+" terminal



$$\frac{v_+ - V_{REF}}{R_1} + \frac{v_+ - v_I}{R_2} = 0$$

$$\Rightarrow \boxed{\begin{array}{l} v_+ = V_{REF}\left(\frac{R_2}{R_1 + R_2}\right) + v_I\left(\frac{R_1}{R_1 + R_2}\right) \\ v_- = 0 \end{array}}$$

$v_+ > v_-$

$\Rightarrow v_I \frac{R_1}{R_1 + R_2} + V_{REF}\frac{R_2}{R_1 + R_2} > 0$

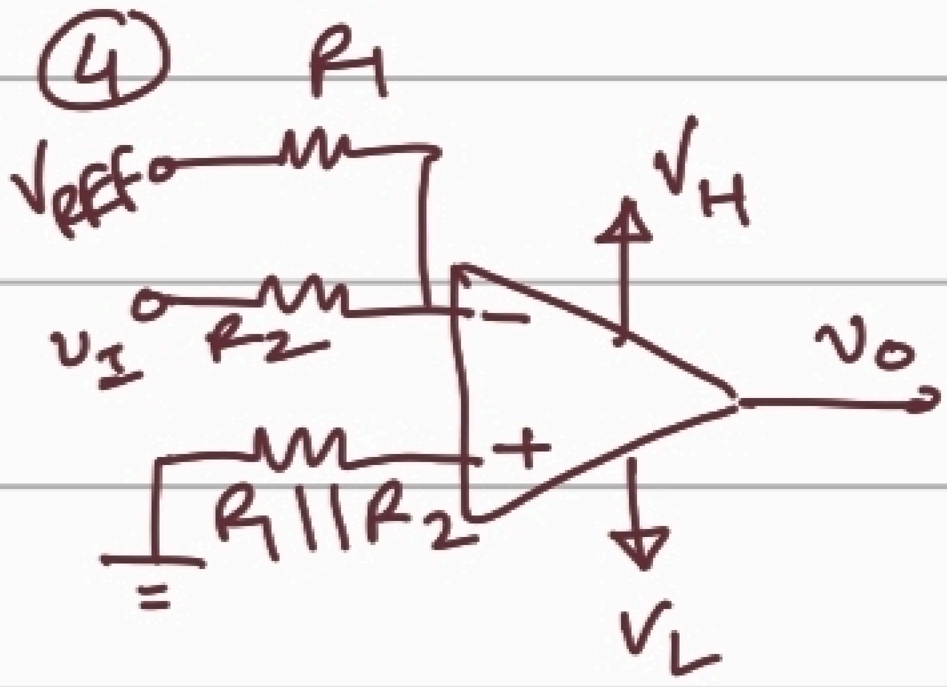
$\Rightarrow v_I > \left(-\frac{R_2}{R_1}\right) V_{REF}$

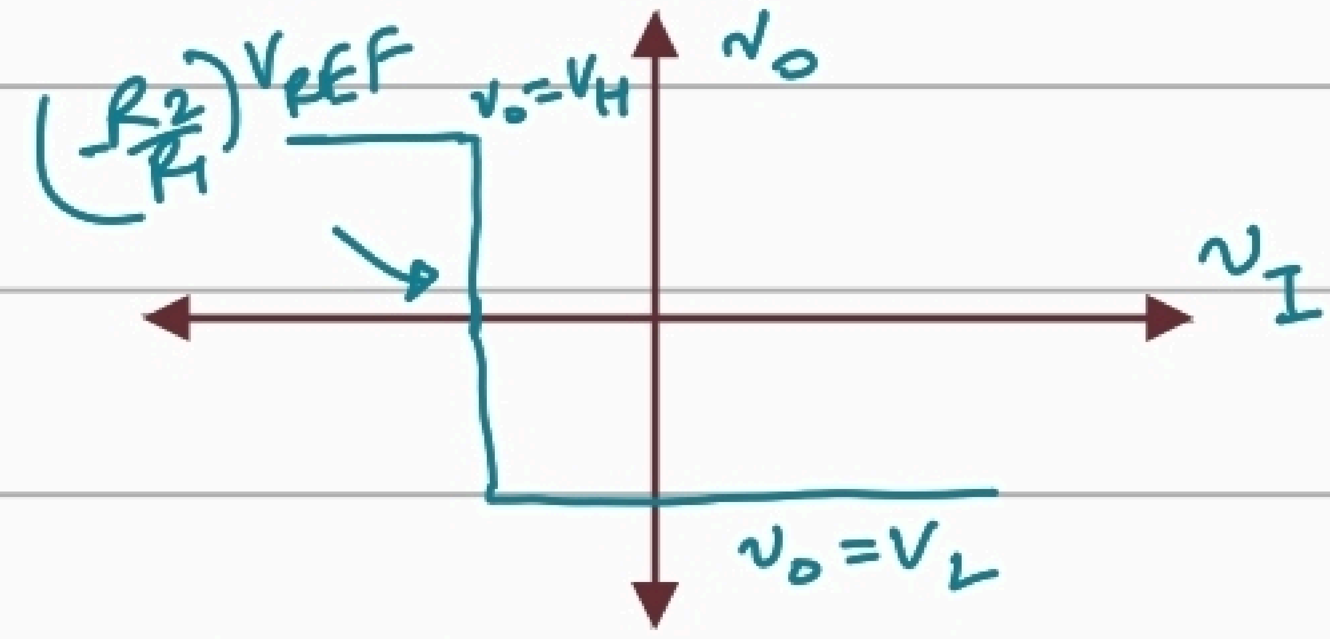
$\Rightarrow v_o = V_H$

or $v_I < \left(-\frac{R_2}{R_1}\right) V_{REF}$

$\Rightarrow v_o = V_L$

(4)





$v_+ = 0$

and $v_- = v_I \dfrac{R_1}{R_1+R_2} + V_{REF} \dfrac{R_2}{R_1+R_2}$

$v_+ > v_-$

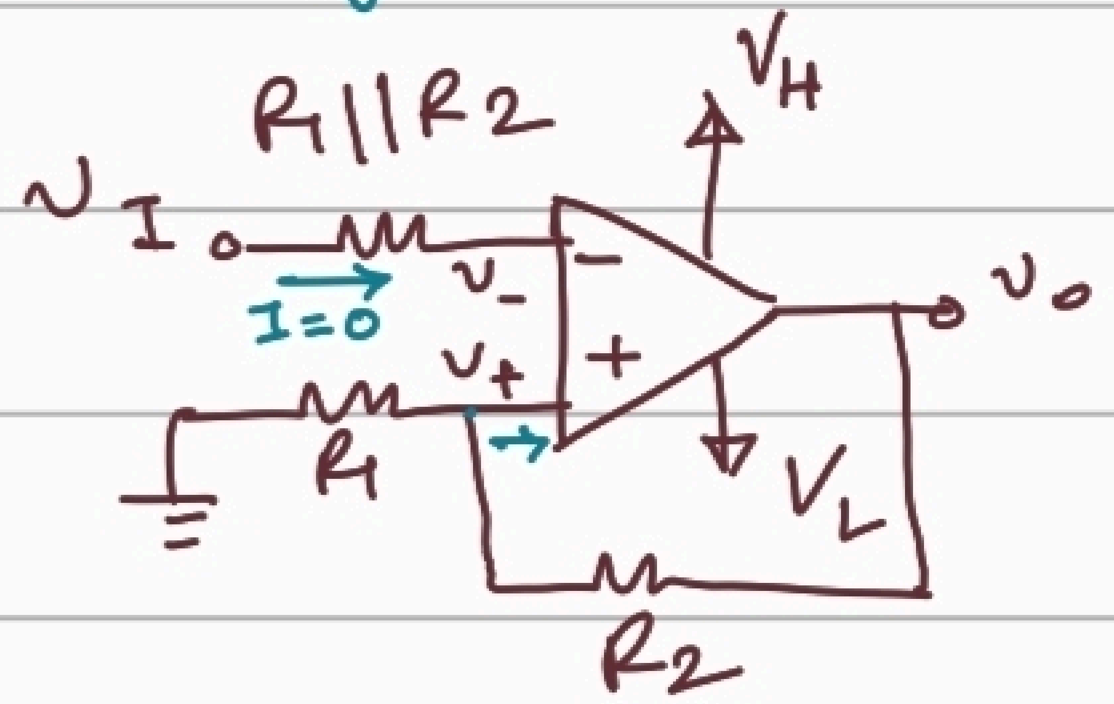
$\Rightarrow v_I < \left(-\frac{R_2}{R_1}\right) V_{REF}$

$\Rightarrow v_o = V_H$

otherwise, $v_o = V_L$

Schmitt Trigger

Inverting S.T.



$$\boxed{\begin{aligned} v_- &= v_I \\ v_+ &= v_o \frac{R_1}{R_1+R_2} \end{aligned}}$$

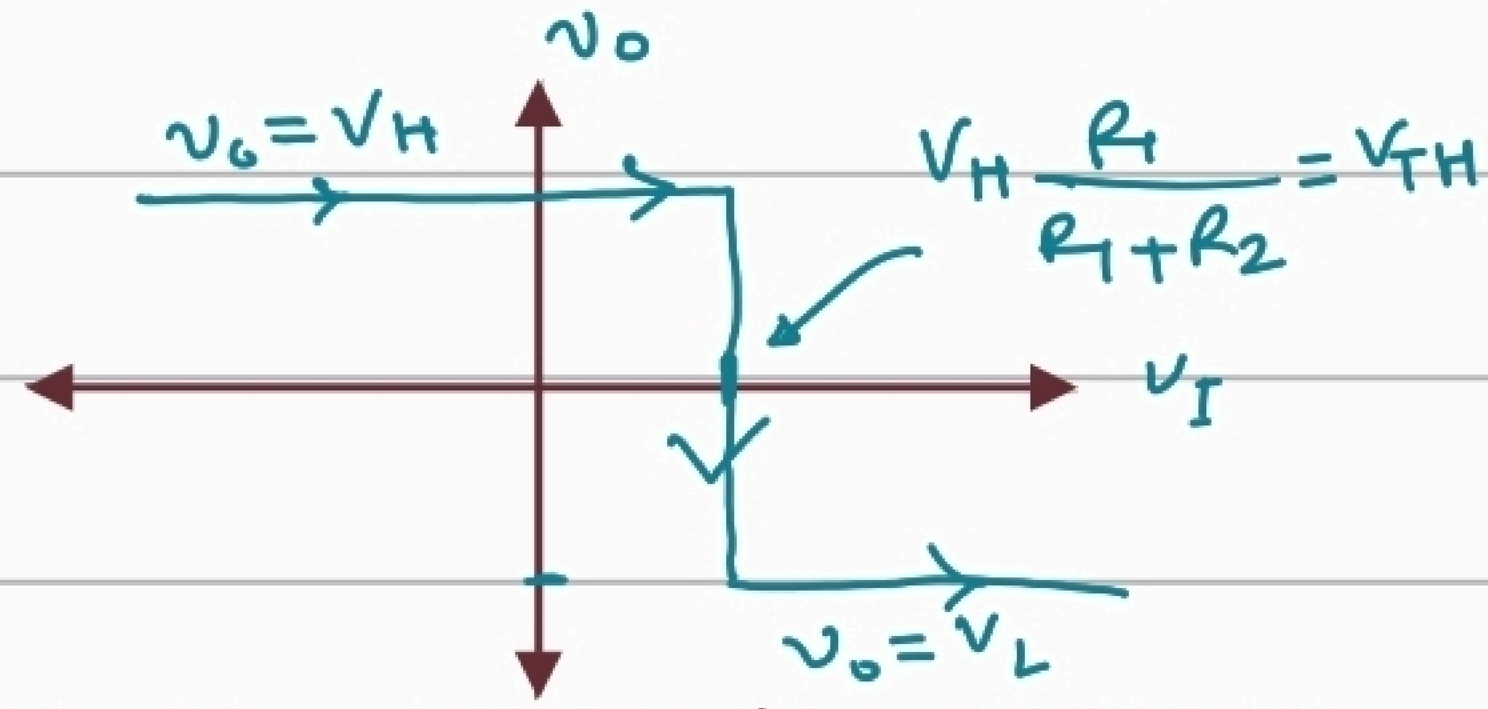
[use node eq^n at "+" terminal]

⊞ Output depends on the initial condition of the input.

Case 1: $|v_I| \gg 0$ but v_I is negative

We can expect, $v_+ > v_-$

$\Rightarrow v_o = V_H$ and $v_+ = V_H \dfrac{R_1}{R_1+R_2}$

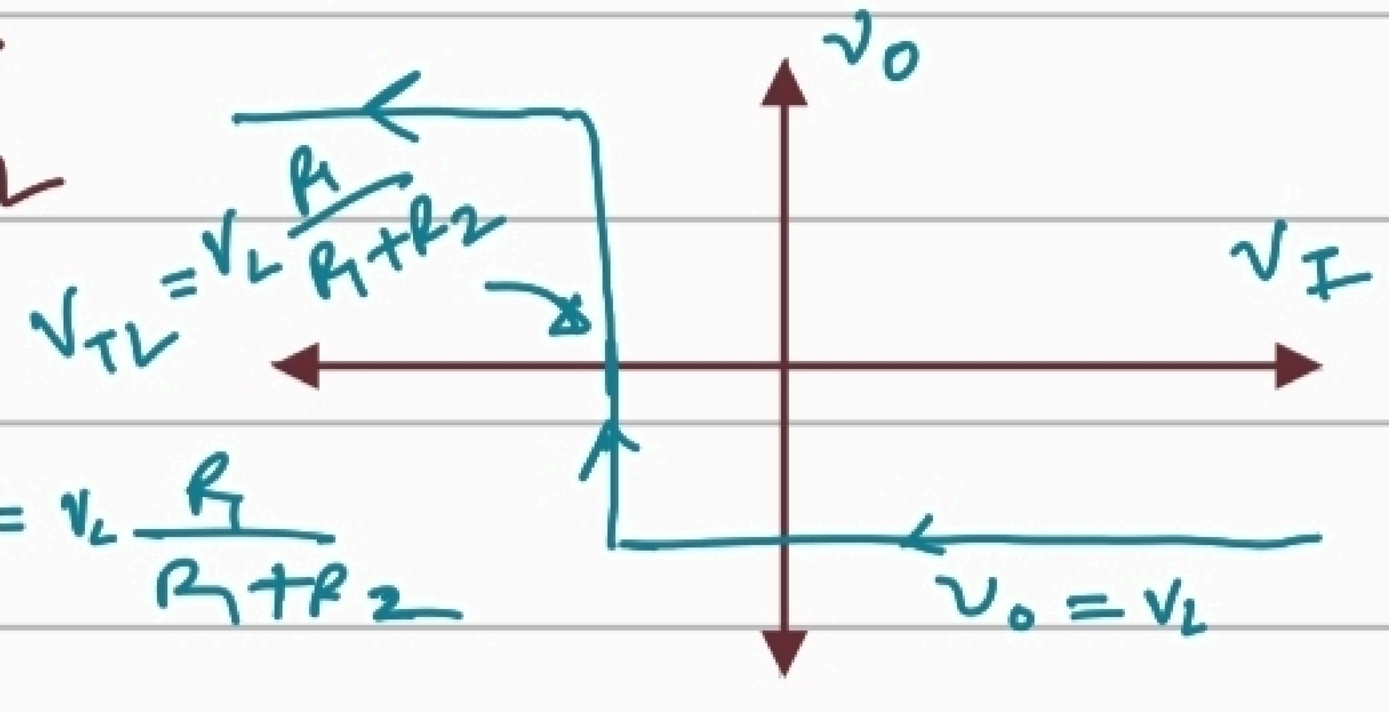


We can increase $v_I = v_-$ until it crosses the voltage of $v_+ = \dfrac{R_1 V_{REF}}{R_1+R_2}$. After that output changes to V_L.

So $V_{REF} \cdot \dfrac{R_1}{R_1+R_2}$ is denoted as higher threshold voltage. V_{TH}

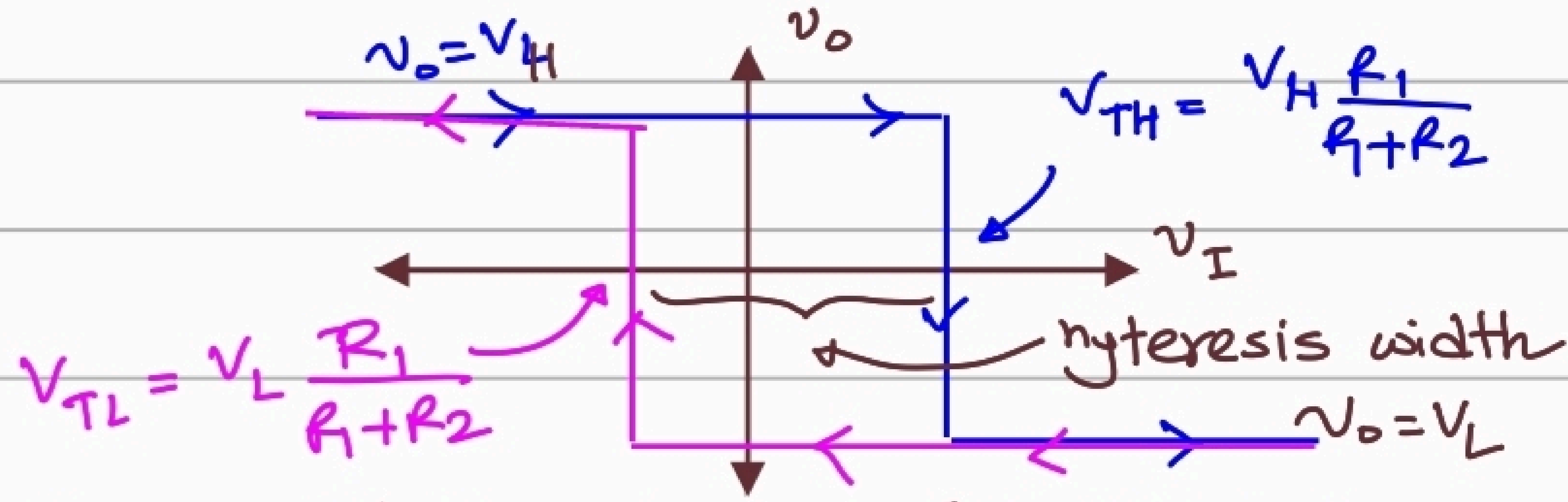
Case 2: $|v_I| \gg 0$ but v_I is positive. We can expect that $v_- > v_+$ and $v_o = V_L \Rightarrow v_+ = V_L \dfrac{R_1}{R_1+R_2}$.

When v_I decreases below the $v_+ = V_L \dfrac{R_1}{R_1+R_2}$, v_o changes from V_L to V_H.



Lower threshold voltage. $V_{TL} = V_L \dfrac{R_1}{R_1+R_2}$

Combined transfer characteristics.



Bistable multivibrator = Schmitt trigger
It has two stable output states. $v_o = V_H / V_L$

Example 15.6 Neamann

Determine the hysteresis width of a particular Schmitt trigger.

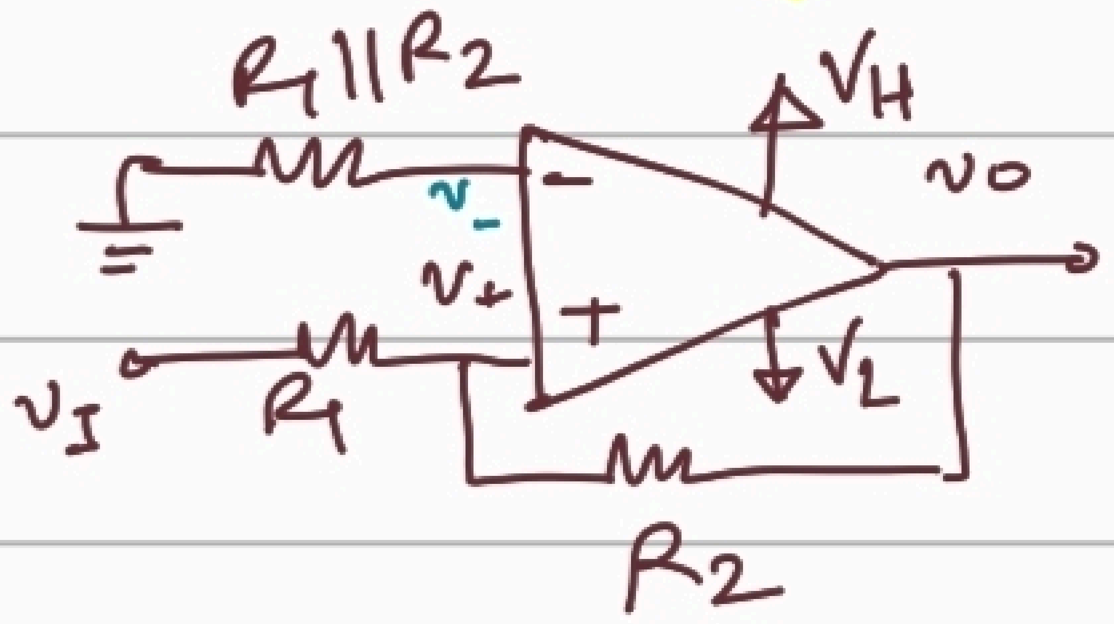
$R_1 = 10k\Omega$, $R_2 = 90k\Omega$, $V_H = 10V$ and $V_L = -10V$.

Sol^n:

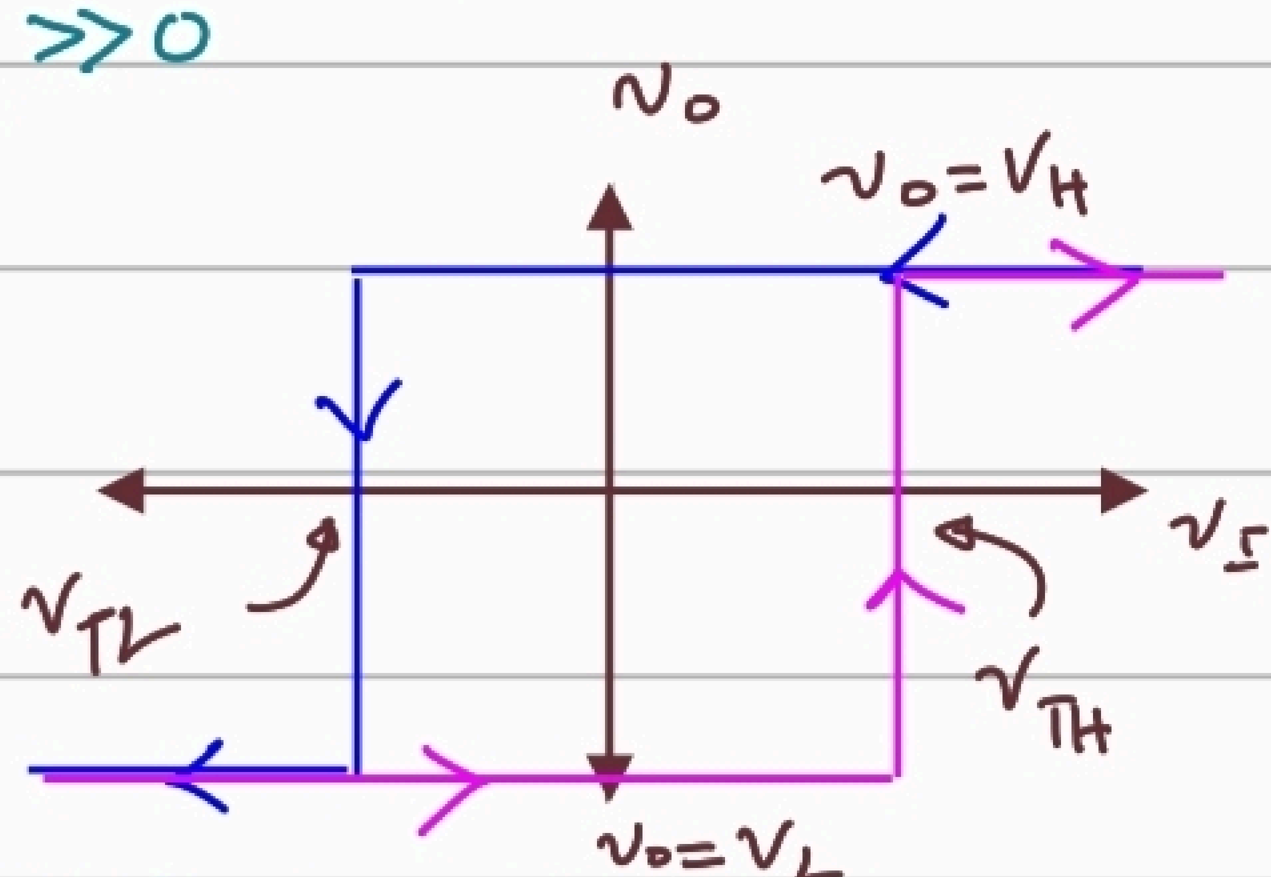
Hysteresis width $= V_{TH} - V_{TL}$

$$\Rightarrow V_{TH} - V_{TL} = V_H \frac{R_1}{R_1+R_2} - V_L \frac{R_1}{R_1+R_2} = (V_H - V_L)\frac{R_1}{R_1+R_2} = 20 \times \frac{10}{100} = 2V$$

Non-inverting Schmitt Trigger Circuit:



Consider the two cases just like before and use $|v_I| >> 0$



$$\boxed{\begin{aligned} v_- &= 0 \\ v_+ &= v_o \frac{R_1}{R_1+R_2} + v_I \frac{R_2}{R_1+R_2} \end{aligned}}$$

At transition point $v_+ = v_-$ and for V_{TH} we can say from transfer characteristics that $v_o = V_L$

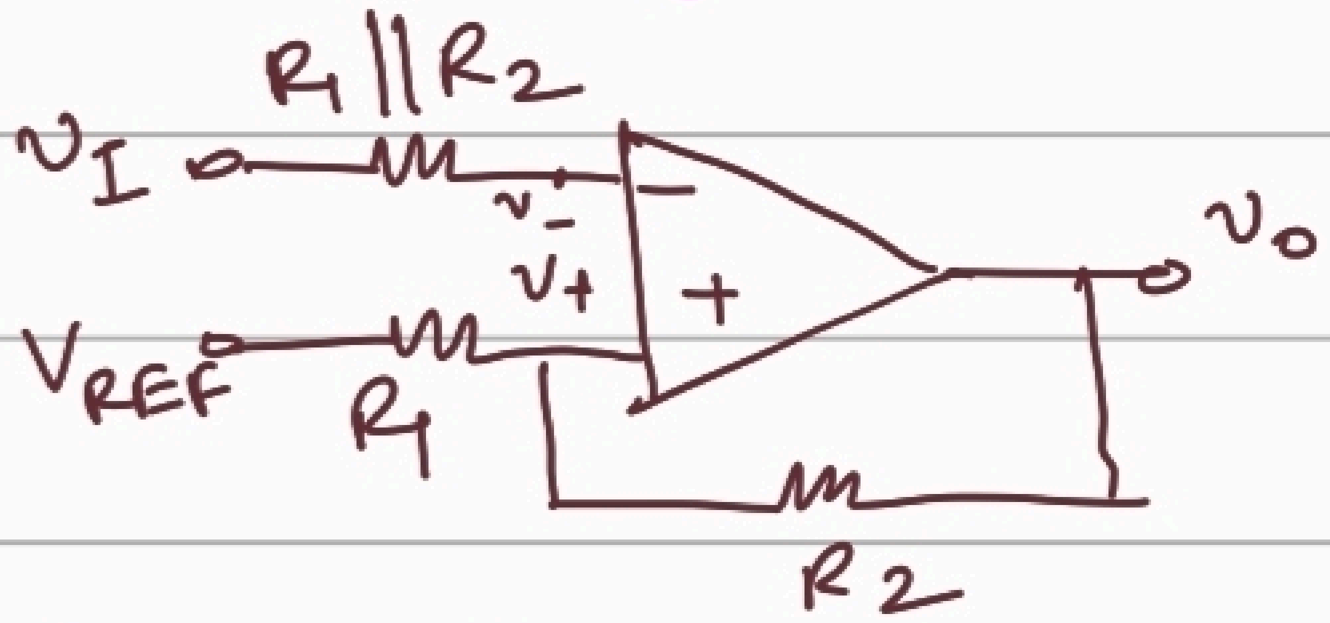
$$0 = V_L \frac{R_1}{R_1+R_2} + V_{TH}\frac{R_2}{R_1+R_2} \Rightarrow \underline{V_{TH}} = \left(-\frac{R_1}{R_2}\right)\underline{V_L}$$

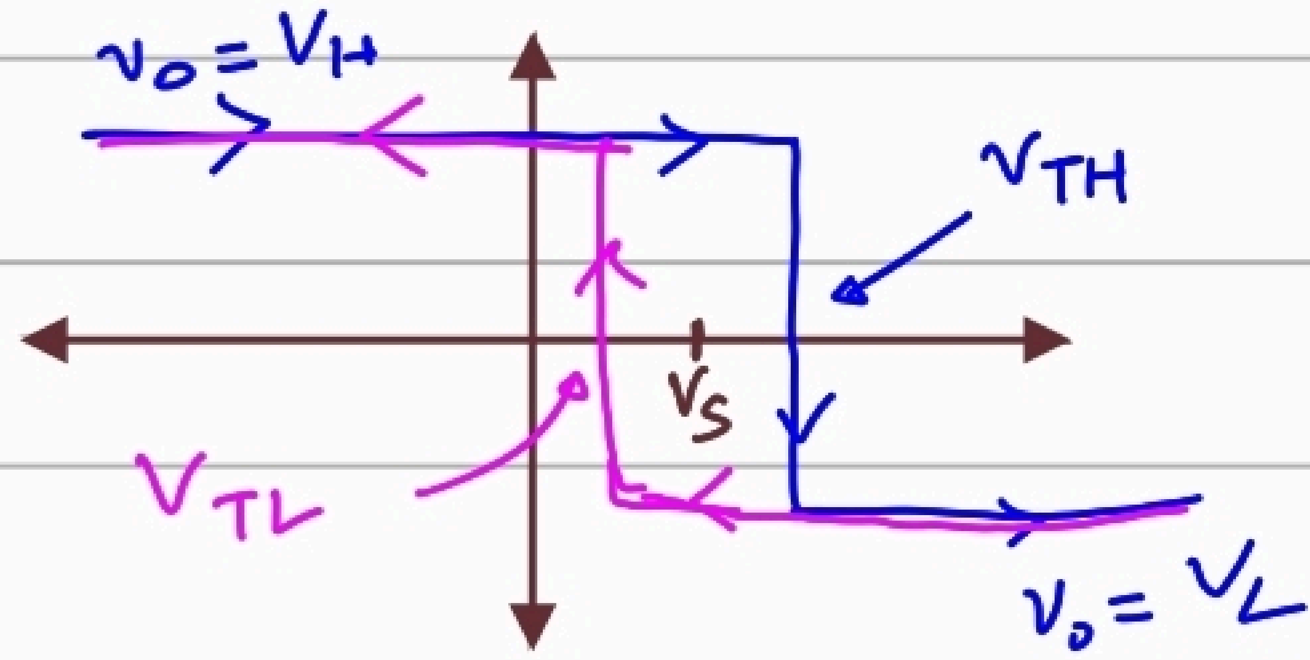
Similarly, we can get. $\underline{V_{TL}} = \left(-\frac{R_1}{R_2}\right)\underline{V_H}$

$v_I \uparrow\downarrow$, $v_o \uparrow\downarrow$ when $|v_I| >> 0$

Schmitt Trigger with Applied Voltage:

Inverting S.T.





V_S = shift voltage.

$$\boxed{\begin{aligned} v_- &= v_I \\ v_+ &= V_{REF}\frac{R_2}{R_1+R_2} + v_o \frac{R_1}{R_1+R_2} \end{aligned}}$$

Now. $v_- = v_+ \; (v_o = V_H,\, V_I = V_{TH})$

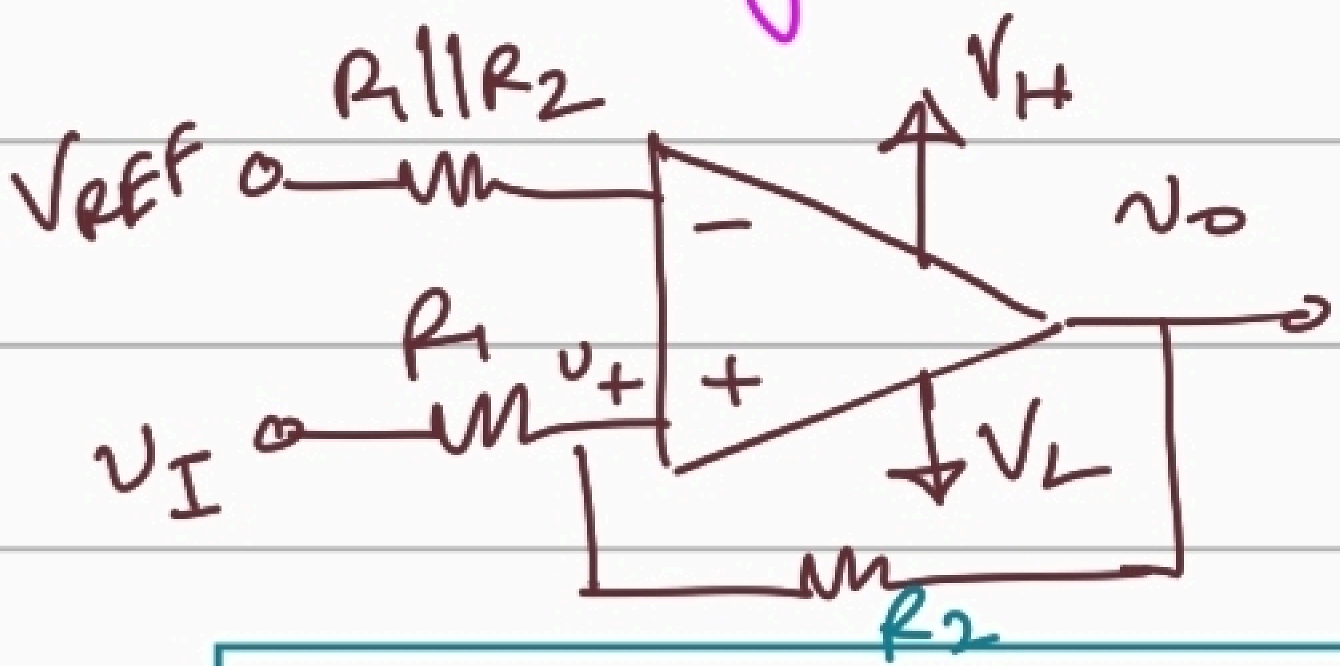
$$\Rightarrow V_{TH} = \underbrace{V_{REF}\left(\frac{R_2}{R_1+R_2}\right)}_{V_S} + V_H\left(\frac{R_1}{R_1+R_2}\right)$$

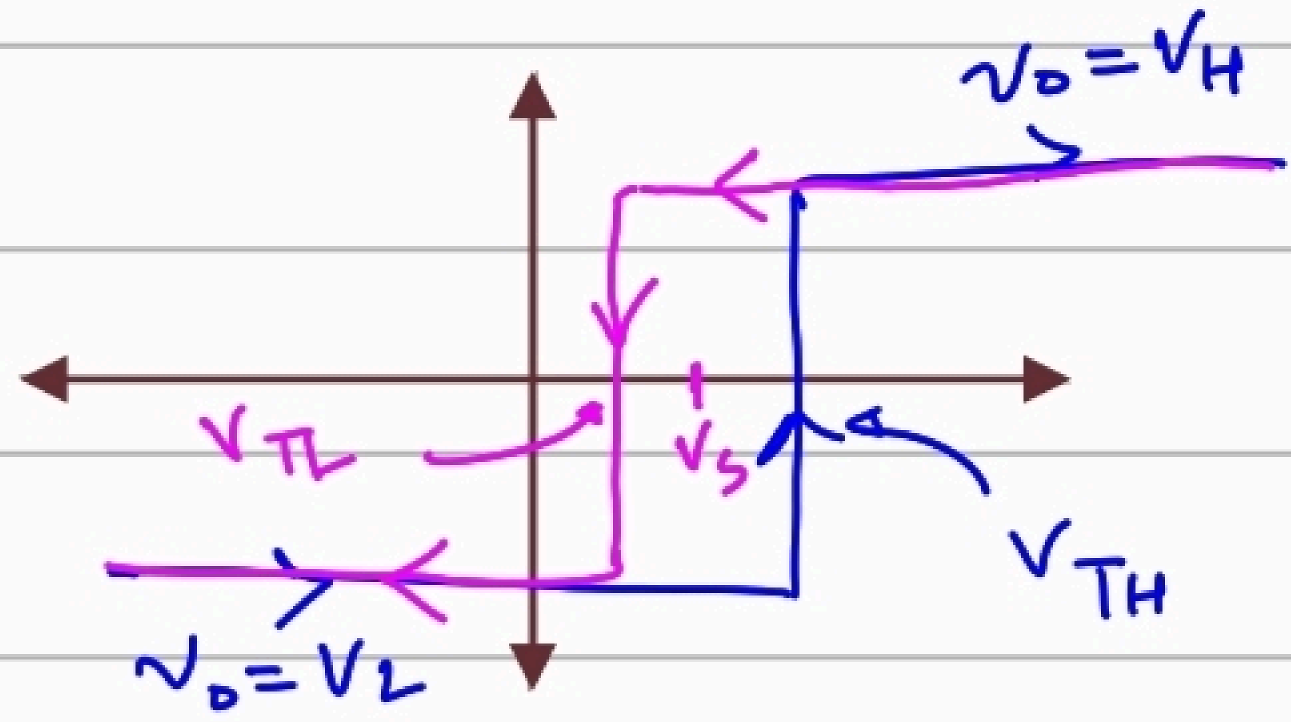
Similarly, $v_- = v_+ \; (v_o = V_L,\, v_I = V_{TL})$

$$\Rightarrow V_{TL} = \underbrace{V_{REF}\left(\frac{R_2}{R_1+R_2}\right)}_{V_S} + V_L\left(\frac{R_1}{R_1+R_2}\right)$$

So shift/switching voltage. $\boxed{V_S = V_{REF}\frac{R_2}{R_1+R_2}}$

NonInverting S.T.





$$\boxed{\begin{aligned} v_- &= V_{REF} \\ v_+ &= v_I \frac{R_2}{R_1+R_2} + v_o \frac{R_1}{R_1+R_2} \end{aligned}}$$

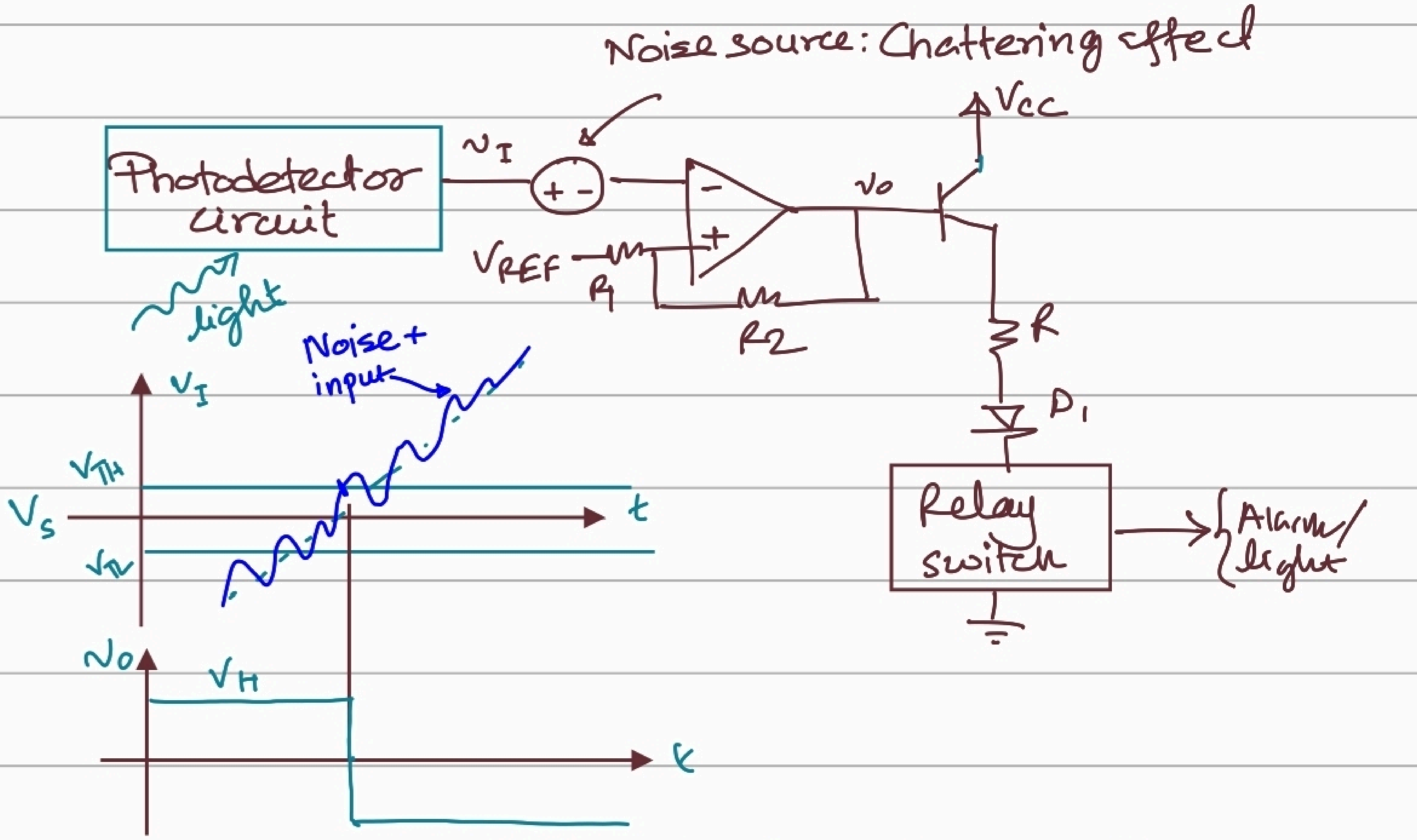
$$V_{TH} = \left(\frac{R_1+R_2}{R_2}\right)V_{REF} + \left(-\frac{R_1}{R_2}\right)V_L$$

$$V_{TL} = \left(\frac{R_1+R_2}{R_2}\right)V_{REF} + \left(-\frac{R_1}{R_2}\right)V_H$$

Shift/switching voltage. $V_S = \left(\frac{R_1+R_2}{R_2}\right)V_{REF}$

Application:

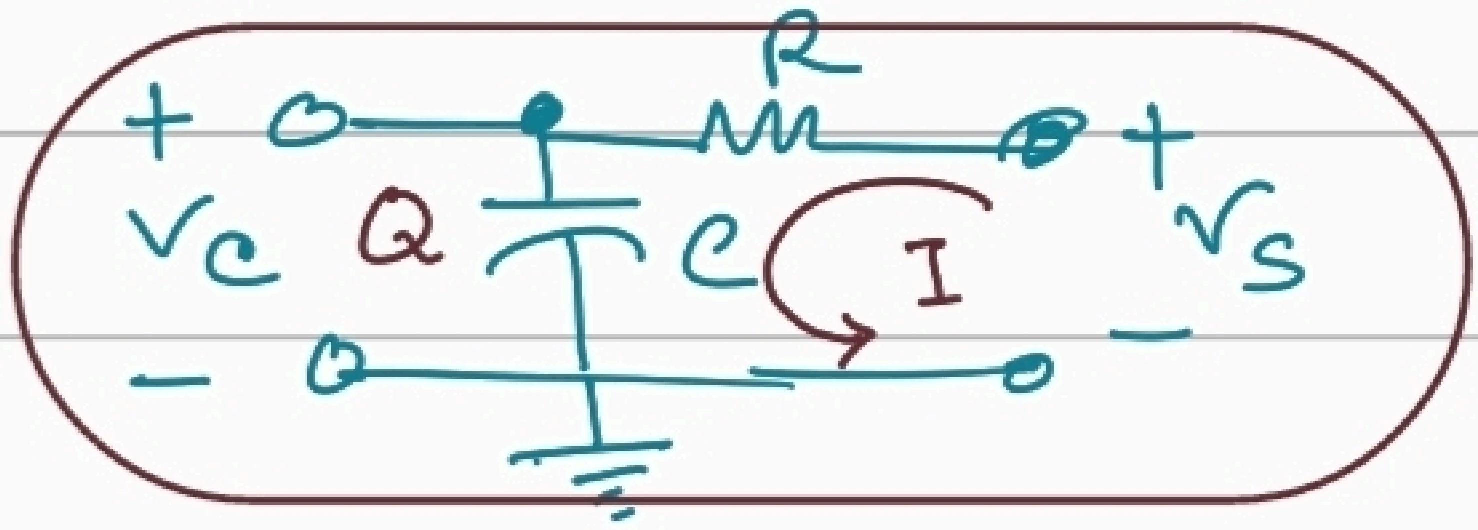
- ☐ Smoke Detector
- ☐ Street light Control



Design problem 15.7 (Practice | Final Exam)

Lecture 19: Signal Generators.

Basic R-C circuit:



Suppose, at initial time t_1 the capacitor has a voltage $v_C(t_1) = v_{initial}$.

If $v_S > v_C(t_1)$, $Q\uparrow$ and if $v_S < v_C(t_1)$, $Q\downarrow$.

From the definition of current flow, we know that, $I = \frac{dQ}{dt}$

Using the definition of capacitance $\left(C = \frac{Q}{V}\right)$

$$I = \frac{dQ}{dt} = \frac{d}{dt}(CV_C) = \boxed{C\frac{dV_C}{dt} = \frac{V_S - V_C}{R}}$$

We want to determine the voltage across capacitor for the time $t_1 < t < \infty$.